Online Appendix
Constructing the Trademark–CRSP Link Table

This appendix describes the procedure we used to construct the trademark–CRSP link table. The construction proceeds in six stages: (1) compiling the public-firm name universe; (2) preparing the trademark owner name file; (3) fuzzy name matching; (4) building the trademark lifecycle panel; (5) linking trademarks to public firms over time; and (6) adjusting ownership for Mergers and Acquisitions (M&A). The resulting table maps over 1.1 million USPTO trademark serial numbers to the permanent number (PERMNO) of the corresponding Compustat/CRSP public firm. The link table is publicly available on GitHub[1].

I. Compiling the Public-Firm Name Universe

A central challenge in linking trademark records to public firms is that the USPTO does not maintain a unique firm identifier that can identify the ultimate domestic parent in a corporate "family tree". At the trademark application stage, the USPTO requires only that applicants provide the owner's name as a free-text string. A public firm may therefore appear under dozens of variants—including the names of its subsidiaries, abbreviated forms, and historical names. We address this challenge by compiling a comprehensive name universe covering all Compustat/CRSP public firms and their subsidiaries. We then use this name universe to search USPTO records and link trademarks to public firms.

A. Source Datasets

We draw on five name fields from the Wharton Research Data Service (WRDS) suite. For each PERMNO, we extract: (i) *HCONM* (historical company name, available from 2007 onward), (ii) *COMNAM* (current company name), and (iii) *ISSUERNM* (issuer name) from CRSP; (iv) *CONM* (current company name) and (v) *CONML* (company legal name) from Compustat. CRSP name fields are downloaded at monthly frequency; Compustat name fields are downloaded at quarterly frequency. We supplement these with subsidiary names from the Subsidiary Data by WRDS dataset, which is constructed from 10-K and 10-Q filings submitted to the SEC between

[1] https://github.com/Shiheli1992/Trademark_CRSP

1995 and 2019 and records the names of all entities listed as subsidiaries in annual and quarterly reports.

B. GVKEY–PERMNO Panel

We construct a monthly GVKEY–PERMNO panel using the CRSP–Compustat Matching (CCM) table. Using the *LINKDT* and *LINKENDDT* fields, we generate a monthly date sequence covering the full valid-link window, expressed as end-of-month dates. Missing *LINKENDDT* values are set to the current date when the table was generated. We then merge all name fields downloaded in Section I.A onto this time-varying GVKEY–PERMNO panel. Table OA1 shows the structure of the panel for McDonald's Corp in January 2007. The name fields *CONM*, *CONML*, and *ISSUERNM* take distinct values across sources, and the last three columns contain duplicated entries across rows. The name stacking and deduplication step described in Section I.C below resolves these redundancies.

Table OA1: GVKEY-PERMNO Panel (McDonald's Corp in January 2007)

GVKEY	LPERMNO	Date	CONM	CONML	ISSUERNM	COMNAM	HCONM
7154	43449	31/01/2007	MCDONALD'S CORP	McDonald's Corp	MCDONALDS CORP	MCDONALDS CORP	MCDONALD'S CORP

We supplement the name universe with subsidiary names by merging them into the GVKEY–PERMNO panel constructed above, matching each subsidiary's report date to the panel date within a ±10-day window. We retain only subsidiaries reported in 10-K and 10-Q filings, and drop records with missing or empty name fields or a GVKEY equal to zero. Where a subsidiary name is associated with more than one GVKEY in the same reporting period, we exclude the observation to avoid ambiguous matches.

C. Name Stacking and Standardization

We apply a uniform cleaning pipeline to all names in the GVKEY–PERMNO panel (Bessen, 2009; Autor et al., 2020). The pipeline proceeds in the following steps:

- Convert all characters to upper case.
- Remove periods, commas, and other non-alphanumeric characters.

- Apply a standardization dictionary of approximately 60 common legal-form suffixes and industry terms, replacing each with a canonical abbreviated form using word-boundary matching (e.g., *CORPORATION → CORP.*; *HOLDINGS → HLDG.*; *INTERNATIONAL → INTL.*; *TECHNOLOGIES → TECH.*). The full replacement table covers entity types (Corp., Inc., Ltd., LLC, PLC, NV, SA, AG, AB), descriptive terms (Holdings, Group, Services, Products, Industries, Management, Communications, Manufacturing), and common abbreviations in multiple forms. This standardization approach follows the practice established in the patent-to-firm matching literature.

- Remove any remaining special characters introduced or retained after dictionary substitution, including apostrophes, percent signs, dollar signs, hash marks, parentheses, brackets, slashes, question marks, exclamation marks, plus signs, colons, and semicolons.

- Collapse consecutive single capital letters separated by spaces (e.g., *A A R → AAR*).

- Trim leading and trailing whitespace and collapse internal multi-spaces to single spaces.

- Convert all characters back to lower case.

After deduplication, the cleaned name universe contains approximately 795,000 unique name strings, each associated with at least one PERMNO. Each cleaned name is then assigned a unique numeric key that serves as the identifier in the subsequent matching stage. Table OA2 illustrates the name universe for McDonald's Corp after these steps. We identify 173 unique names associated with McDonald's Corp, the majority of which are subsidiary names.

Table OA2: Name Universe -- McDonald's Corp (randomly selected 10 rows)

LPERMNO	Clean_Name	CCM_Name_Key
43449	mcd luxembourg hldg sa rl	236829
43449	mcdi hldg	236830
43449	mcdonalds amea llc	236831
43449	mcdonalds apmea llc	236832
43449	mcdonalds australia	236833
43449	mcdonalds australia hldg	236834
43449	mcdonalds australia ltd	236835
43449	mcdonalds australian property corp	236836
43449	mcdonalds australian property funding llc	236837
43449	mcdonalds beteiligungs gmbh	236838

II. Preparing the Trademark Owner Name File

To construct the trademark owner name universe for the fuzzy matching procedure, we download the *owner.csv* file directly from the USPTO Trademark Case Files Dataset (Graham et al. 2013). The file records the name and type of each trademark owner at each stage of the trademark's lifecycle. We first deduplicate owner names within each serial number, and then apply the same name-cleaning pipeline described in Section I.C to the *own_name* field of each trademark owner record. Each distinct cleaned owner name is assigned a unique numeric identifier used in the subsequent matching stage.

III. Fuzzy Name Matching

We match trademark owner names to public-firm names using a two-stage procedure. We first use TF-IDF vectorization to generate candidate matches and then apply Jaro–Winkler similarity to rank and filter those candidates. Cohen, Ravikumar, and Fienberg (2003) show that this combination outperforms either method alone on name-matching tasks, and our procedure closely follows the trademark-to-public-firm matching approach of Heath and Mace (2020).

A. Stage 1: TF-IDF Candidate Generation

We use the PolyFuzz library in Python, which implements TF-IDF vectorization over character-level tokens[2]. For each trademark owner name, the algorithm returns the top 50 candidate public-firm names ranked by cosine similarity of their TF-IDF vectors. The output is a candidate table of (trademark owner name, public-firm name, TF-IDF score) triples, yielding up to 50 candidates per trademark owner name.

B. Stage 2: Jaro-Winkler Scoring and Threshold Selection

For each candidate pair in the TF-IDF output, we compute a Jaro–Winkler similarity score using the *textdistance* Python library[3]. We retain all candidate pairs with a score above 0.97 for

[2] The TF-IDF approach has a well-established advantage over pure edit-distance methods in firm-name matching. It down-weights tokens that appear frequently across the entire corpus (such as common legal suffixes) and up-weights tokens that are distinctive. As a result, two strings that share a rare token (e.g., a specific brand term) receive a higher similarity score than two strings that share only a common suffix (e.g., CORP.). This property is particularly important in the trademark setting, where most names in both datasets contain the same small set of legal-form abbreviations after standardization. Cohen, Ravikumar, and Fienberg (2003) demonstrate empirically that a hybrid TF-IDF/Jaro–Winkler scheme outperforms either method alone across a range of name-matching benchmarks.

[3] We use Jaro–Winkler similarity because it is specifically designed for short name strings and accommodates the transpositions and prefix variations that are common in our setting. The metric was originally developed by Jaro (1989) and extended by Winkler (1990) to give additional weight to strings sharing a common prefix, which is particularly apt for company-name variants.

manual inspection. We select this threshold based on preliminary validation, which shows that credible matches rarely fall below this value, as reported in Table OA3. We additionally exclude pairs whose names are shorter than six characters and whose similarity score equals 1.00, as very short strings that match exactly are often common words or abbreviations shared by unrelated entities and cannot be reliably verified as true matches.

Table OA3: Manual inspection for setting the threshold

Row Number	# of Row checked	Matched	Similarity Score	Matching Rate
1-101	100	99	0.995238	99.00%
3000-3103	104	104	0.990000	100.00%
9998-10101	103	11	0.983333	10.68%
16272-16375	104	41	0.979487	39.42%
20000-20100	101	4	0.977778	3.96%
40000-40100	101	0	0.970833	0.00%
40101-40200	100	4	0.970833	4.00%
40201-40300	99	0	0.970833	0.00%
40301-40500	200	2	0.970833	1.00%

C. Manual Verification

Each candidate pair above the 0.97 threshold is manually reviewed. We retain only pairs for which we are confident that the trademark owner name and the public-firm name refer to the same underlying legal entity. Uncertain, ambiguous, and clearly incorrect matches are discarded. The resulting verified mapping table records, for each trademark owner name, the associated PERMNO and the matched public-firm name. Table OA4 presents an example. This mapping table is used in Section V to link trademark records to public firms.

Table OA4: Verified Mapping Table – McDonald's Corp

Trademark Owner Name	Public Firm Name	LPERMNO	Similarity Score
boston market corp	boston market corp	43449	1.0000
chipotle mexican grill inc	chipotle mexican grill inc	43449	1.0000
chipotle mexican grill inc	chipotle mexican grill inc	43449	1.0000
mcdoanlds corp	mcdonalds corp	43449	0.9857
mcdonald corp	mcdonalds corp	43449	0.9857
mcdonald s corp	mcdonalds corp	43449	0.9724
mcdonalds corp	mcdonalds corp	43449	1.0000
mcdonalds intl property com ltd	mcdonalds intl property com ltd	43449	1.0000
mcdonaldss corp	mcdonalds corp	43449	0.9867

IV. Building the Trademark Ownership Table

A second challenge is that the USPTO Trademark Case File does not record ownership structure over time. Each trademark is associated with an owner, but no timestamps indicate when that ownership began or ended. To identify the alive trademarks held by a public firm at any given point in time, we construct a trademark ownership table with dated ownership periods in two steps. First, we complete the trademark date fields using the *event.csv* file from the USPTO Trademark Case Files Dataset, establishing the lifespan of each trademark from application to expiration. We then reconstruct ownership histories over time using the USPTO Trademark Assignment Dataset.

A1. Completing Trademark Case File Dates

The *case_file.csv* file under USPTO Trademark Case File contains key lifecycle dates — filing, publication, and registration — but these fields contain missing values. We supplement missing dates using the *event.csv* file from the USPTO Trademark Case Files Dataset, which records a timestamped event for each status change in a trademark's prosecution history[4]. This step yields the starting date for each trademark.

A2. Determining Trademark Expiration Dates

We next identify the ending date for each trademark. A trademark's validity period ends upon cancellation, abandonment, or expiration of its registration term. We construct an expiration date for each trademark following a priority hierarchy:

1. If a cancellation date (*reg_cancel_dt*) is present in *case_file.csv*, we use it as the end date.
2. If no cancellation date is present but an abandonment date (*abandon_dt*) is available in *case_file.csv*, we use the abandonment date.
3. If neither cancellation nor abandonment is recorded but renewal activity is present, we determine the end date from the most recent renewal event. Where a renewal date (*renewal_dt*) is recorded directly in *case_file.csv*, we use it as the primary source; we further complete renewal histories using the *event.csv* file. The USPTO's renewal system has two regimes: renewals recorded before November 16, 1989 carry a 20-year term;

[4] For each event type, we extract the earliest event date per serial number and use it to populate the corresponding date field in the Case File. The relevant event codes are: *NWAP* and *NWOS* for new applications; *NPUB* for publication; and *NRCC* and *NRCS* for registration.

renewals recorded on or after that date carry a 10-year term. Accordingly, for each trademark, we identify the latest renewal date across all renewal event codes (*REN1–REN5* for 20-year renewals; *RNL1–RNL9*, *8.OK*, *8.PR*, *C15A*, and *C15P* for 10-year renewals) and set the end date to the renewal date plus the applicable term.

4. If no renewal activity is present but a registration date is available, we assign expiration as the registration date plus 20 years (if registered before November 16, 1989) or 10 years (if registered on or after that date), following the change in statutory term length.
5. For trademarks that were never registered, we use the publication date as the end date if available, or the filing date otherwise.

Trademarks with a computed end date beyond December 31, 2024 are assigned a placeholder end date of December 31, 2999 to indicate that the mark remains alive at the close of the sample period.

Table OA5 illustrates how we determine the ending date for a trademark under different conditions. In the first row, the trademark was renewed under the code *RNL2* (from the *event.csv* file), and it remains an alive trademark, as it is still within the 10-year validity period following the latest renewal date. Accordingly, the ending date is assigned December 31, 2999, to indicate that the mark remains alive at the time we generate the dataset. The second trademark expired on February 25, 2006: renewal activity is present, and the latest renewal predates November 16, 1989, so the trademark had a 20-year validity period from that renewal date. The third trademark would nominally expire on January 17, 2019 — ten years after the 2009 renewal — because no subsequent renewal is recorded. However, the *case_file.csv* records a cancellation on October 15, 2021; since *case_file.csv* cancellation takes priority under our hierarchy, we set the end date as recorded in the *case_file.csv* file.

As shown in the first three rows of Table OA5, each contains a renewal code, indicating that the renewal dates were populated by tracing prosecution history and documents in the *event.csv* file. In contrast, the last row records a renewal date directly in the original *case_file.csv* file under the *renewal_dt* column. This trademark expired 10 years after that renewal because no subsequent renewal is recorded.

Table OA5: Trademark End Date Determined under Different Conditions

Serial No	Filing Date	Reg Cancel Date	Renewal Date	Renewal Code	Renew Date	Registration Date	Expired Date
70013064	08/02/1886			RNL2	2/03/2016	02/03/1886	31/12/2999
70027864	24/01/1896			REN4	25/02/1986	25/02/1896	25/02/2006
70032301	18/11/1898	15/10/2021		RNL1	17/01/2009	27/12/1898	15/10/2021
71120899	24/07/1919		18/05/2000		18/05/2000	18/05/1920	18/05/2010

B. Reconstructing Ownership Periods from Assignment Records

The USPTO Trademark Assignment Dataset records each transfer of trademark ownership (Graham, Marco, and Myers 2018). We use it to construct a complete sequence of ownership periods for each trademark. The assignment file is assembled from five component tables: *tm_assignment* (transaction metadata including record date and conveyance text), *tm_docid* (mapping assignment transactions to serial numbers), *tm_assignee* (names of receiving parties), *tm_assignor* (names of transferring parties), and *tm_convey* (conveyance type classification). We exclude transactions classified as “security interest” or “release”, as these represent collateral arrangements rather than changes in beneficial ownership.

Assignee names in Trademark Assignment Dataset could differ from the corresponding owner names in the *owner.csv* file from Trademark Case Dataset. We therefore apply the Jaro–Winkler algorithm to fuzzy-match owner names across the two sources. For each (serial number, owner name) pair in the *owner.csv* file, we compute the Jaro–Winkler distance to all assignee names associated with the same serial number in the *tm_assignee.csv* file, and retain the best match per owner name with a similarity score above 0.95. We then manually verify these pairs and construct a final matching table linking owner names in the USPTO Trademark Case File Dataset to assignee names in the USPTO Trademark Assignment Dataset.

Owner type codes in the *owner.csv* file in the USPTO Trademark Case File Dataset indicate the stage at which an owner was recorded: codes beginning with 1 correspond to filing-stage owners, codes beginning with 2 to publication-stage owners, codes beginning with 3 to registrant-stage owners, and codes beginning with 4 to renewal-stage owners. For owners at the filing, publication, and registration stages, we assign the corresponding lifecycle date (*filing_dt*, *publication_dt*, or *registration_dt*) as the ownership start date when no assignment record date is available.

Renewal-stage owners require additional steps. For each trademark, we collect all available renewal dates from the renewal date columns (*REN1–REN5*, *RNL1–RNL9*, *renewal_dt*, *8.OK*, *8.PR*, *C15A*, *C15P*) and rank them chronologically. We then rank renewal-stage owners by their owner type code — where 41 identifies the first renewal owner, 42 the second, and so on — and assign each the renewal date of the corresponding rank. We apply this rank-matching procedure only when the number of renewal-stage owners does not exceed the number of available renewal dates; observations that do not satisfy this condition are dropped, as they imply more owners identified in the USPTO Trademark Assignment Dataset than recorded owners in the *owner.csv* file, making the assignment ambiguous and prone to false positive matches.

Ownership periods are then sorted by serial number and start date. The end date of each ownership period is set to one day before the next owner's start date within the same serial number. The final ownership period for each trademark is terminated at the trademark's computed expiration date. Trademarks with no valid start date are dropped. Table OA6 shows the final data structure of the trademark ownership table. The first three rows show the ownership history for an active trademark with serial no. *70011210*. The last three rows show a trademark with three ownership spells that expired on October 15, 2021, as no renewal is recorded.

Table OA6: Trademark Ownership Table — Illustrative Example

Serial No.	Owner Name	Ownership Start Date	Ownership End Date	Trademark Expired Date
70011210	tolman jampes p	27/05/1884	23/05/1976	
70011210	samson ocean systems inc	24/05/1976	4/03/2003	
70011210	samson rope tech inc	5/03/2003	31/12/2999	
70017953	worcester royal porcelain com ltd	27/05/1890	15/02/1984	15/10/2021
70017953	royal worcester spode ltd	16/02/1984	29/10/2000	15/10/2021
70017953	porcelain and fine china companies ltd	30/10/2000	15/10/2021	15/10/2021

V. Linking Trademarks to Public Firms

Using the mapping table from Section III, we assign each dated trademark–owner pair to a PERMNO. Specifically, we merge the dated ownership panel with the mapping table on the cleaned owner name. We then match each trademark–PERMNO pair to the GVKEY–PERMNO link from the CCM linking table and enforce temporal validity. We assign a trademark to a GVKEY only when the ownership start date (Ownership Start Date in Table OA6) falls strictly within the CRSP–Compustat link window, i.e., *LINKDT* < Ownership Start Date < *LINKENDDT*.

VI. Adjusting for Mergers and Acquisitions

Trademark ownership does not always transfer in USPTO records at the time of a corporate acquisition, and assignment filings may be delayed or omitted. To ensure that trademark portfolios are attributed to the correct owner following a change of control, we refine ownership records using deal-level M&A data from SDC Platinum, following the dynamic reassignment approach of Arora, Belenzon, and Sheer (2021). We restrict to completed deals in which the acquirer obtains more than 50% voting control (*pctown* > 50), and identify the effective date (*dateeff*) as the date of ownership transfer.

A. Public Target Acquired by a Public Firm

For trademarks held by a public target at the time of acquisition, we reassign ownership to the acquirer's PERMNO as of the deal effective date. For each target–acquirer pair, we retain only the first deal that conveys controlling rights. We exclude buyback transactions (cases where the acquirer PERMNO equals the target PERMNO). The acquirer's ownership period begins on the deal effective date and extends through the trademark's expiration date. The original owner's ownership period is truncated to end on the day before the effective date of the deal.

B. Public Target Acquired by a Private Firm

When the acquirer is a private entity (no acquirer PERMNO), we cannot reassign ownership to a public firm. Instead, we truncate the target's ownership period at the deal effective date, removing the trademark from the target's portfolio as of that date.

Table OA7 illustrates the dynamic ownership adjustment using the acquisition of 21st Century Fox by Walt Disney Co, which became effective on March 20, 2019. The first three rows show a trademark with two prior owners, the second of which is 21st Century Fox. The Fox ownership period ends on the day before the deal effective date, and Walt Disney Co's ownership period begins the following day. Since no renewal activity is recorded for this trademark, it expired on March 20, 2020, at which point the Walt Disney Co ownership period also ends. The next three rows show an otherwise identical ownership transition, except that the trademark has been renewed and therefore remains alive under Walt Disney Co. The last row illustrates the special case where Fox became the recorded owner of a trademark after the deal effective date. Since beneficial ownership had already transferred to Walt Disney Co, we attribute this trademark directly to Walt Disney Co from the original filing date.

Table OA7: Dynamic Trademark Ownership Adjustment — Walt Disney Co Acquisition of 21st Century Fox

Serial No.	Target Company	Acquirer Company	Date effective	LPERMNO	Ownership Start Date	Ownership End Date	Trademark Expiration Date
73510999				10720	29/11/1984	4/05/2005	20/03/2020
73510999				90441	5/05/2005	20/03/2019	20/03/2020
73510999	21st Century Fox Inc	Walt Disney Co	20/03/2019	26403	21/03/2019	20/03/2020	20/03/2020
73531782				84145	12/04/1985	6/04/2005	31/12/2999
73531782				90441	7/04/2005	20/03/2019	31/12/2999
73531782	21st Century Fox Inc	Walt Disney Co	20/03/2019	26403	21/03/2019	31/12/2999	31/12/2999
73781174	21st Century Fox Inc	Walt Disney Co	20/03/2019	26403	30/07/2019	31/12/2999	31/12/2999

VII. Final Dataset

The final link table contains the following variables for each trademark–owner observation: USPTO serial number, CRSP PERMNO, filing date, registration date, ownership start date, ownership end date, trademark expiration date, and a flag indicating whether the trademark was reassigned to a new PERMNO as a result of the M&A adjustment. Table OA8 illustrates the data structure.

Table OA.8 Final Link Table — Data Structure

LPERMNO	Serial No.	Filing Date	Registration Date	Ownership Start Date	Ownership End Date	Trademark Expiration Date	M&A Buying
13598	71000233	3/04/1905	4/07/1905	10/10/2012	3/07/2015	31/12/2999	FALSE
13901	71000233	3/04/1905	4/07/1905	17/01/1995	9/10/2012	31/12/2999	FALSE
89006	71000233	3/04/1905	4/07/1905	4/07/2015	31/12/2999	31/12/2999	FALSE
26403	73781174	17/02/1989	17/10/1989	30/07/2019	31/12/2999	31/12/2999	TRUE

Table OA9 summarises the final Trademark–CRSP link table. The table contains approximately 1.4 million observations identifying ownership for 1.11 million unique trademarks held by 14,786 public firms, covering the period from October 1870 to March 2024.

Table OA.9 Summary of the Trademark–CRSP Link Table

Data	N
Trademark–owner–year	1,404,605
Unique trademark serial numbers	1,112,213
Distinct public firms (PERMNO)	14,786
Sample period	Oct. 1870 – Mar. 2024

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